

1 PROBLEM

There are $k \geq 3$ distinct positive integers $a_1, \dots, a_k$ with product P and sum strictly less than $3 \cdot 10^6$. Let g be the largest positive integer that divides each of $(P/a_i) + 1$ for $1 \leq i \leq k$. Over all choices of the a_i and all $k \geq 3$, find the maximum possible value of g .

2 ANSWER

The maximum possible value of g is

$$g = 998285.$$

It is attained, for example, by $k = 3$ and

$$a_1 = 1413, \quad a_2 = 999698, \quad a_3 = 1997983,$$

whose sum is $2\,999\,094 < 3\,000\,000$.

3 NOTATION

Throughout write $b_i = P / a_i = \text{product of all } a_j \text{ with } j \neq i$. By definition g is the greatest common divisor $g = \gcd(b_1 + 1, b_2 + 1, \dots, b_k + 1)$, i.e. g is the largest integer with $g \mid b_i + 1$ for every i . Equivalently

$$b_i \equiv -1 \pmod{g} \quad \text{for all } i. \quad (*)$$

We must maximise this gcd over all admissible families (a_i) , $k \geq 3$, subject to

$$a_1, \dots, a_k \text{ distinct positive integers, } a_1 + \dots + a_k < 3 \cdot 10^6.$$

4 PART 1. STRUCTURE FORCED BY THE DIVISIBILITY (*)

Fix any admissible family and let g be its associated gcd.

Lemma 1 (all a_i are congruent mod g). For each i we have $a_i \cdot b_i = P$. Reducing mod g and using $(*)$, $b_i \equiv -1$, so

$$P = a_i \cdot b_i \equiv a_i \cdot (-1) \equiv -a_i \pmod{g} \quad \text{for every } i.$$

Hence for all i, j : $-a_i \equiv P \equiv -a_j \pmod{g}$, i.e.

$$a_i \equiv a_j \pmod{g}.$$

Thus there is a common residue c with $a_i \equiv c \pmod{g}$ for all i .

Lemma 2 (the residue condition). Since every $a_j \equiv c \pmod{g}$, for each fixed i ,

$$b_i = \text{product over } j \neq i \text{ of } a_j \equiv c^{(k-1)} \pmod{g}.$$

Combining with $(*)$, $c^{(k-1)} \equiv -1 \pmod{g}$. In particular $c \cdot c^{(k-2)} \equiv -1$, so c is invertible mod g , i.e. $\gcd(c, g) = 1$. For $k = 3$ this reads

$$c^2 \equiv -1 \pmod{g}. \quad (R)$$

Lemma 3 (a lower bound for the sum). Let r be the least positive residue of c modulo g . Because $\gcd(c, g) = 1$ and (for any nontrivial case) $g \geq 2$, we have $1 \leq r \leq g - 1$; in particular $r \geq 1$. All a_i are distinct positive integers congruent to $r \pmod{g}$. The k smallest distinct positive integers in this residue class are

$$r, \quad r + g, \quad r + 2g, \quad \dots, \quad r + (k-1)g,$$

so

$$\begin{aligned} a_1 + \dots + a_k &\geq k*r + g * (0 + 1 + \dots + (k-1)) \\ &= k*r + g * C(k,2), \end{aligned}$$

where $C(k,2) = k(k-1)/2$. Since the sum is $< 3*10^6$,

$$k*r + g * C(k,2) < 3*10^6. \quad (S)$$

5 PART 2. $k \geq 4$ IS USELESS

For $k \geq 4$ we have $C(k,2) \geq 6$ and $r \geq 1$, so by (S),

$$6g \leq C(k,2) * g \leq k*r + C(k,2)*g < 3*10^6,$$

hence $g < 500000$. This is far below the value 998285 produced by $k = 3$ in Part 4, so the maximum is attained with $k = 3$. From now on take $k = 3$.

6 PART 3. UPPER BOUND FOR $k = 3$: $g \leq 998285$

Let g be the gcd of an admissible family with $k = 3$, and let $r (\geq 1)$ be as in Lemma 3. From (R), $g \mid c^2 + 1$; reducing c to r gives $r^2 \equiv -1 \pmod{g}$, so

$$g \mid r^2 + 1, \quad \text{hence} \quad r^2 + 1 = g * m \quad \text{for some integer } m \geq 1. \quad (D)$$

In particular $g \leq r^2 + 1$. From (S) with $k = 3$ ($C(3,2) = 3$):

$$3r + 3g < 3*10^6 \quad \Rightarrow \quad r + g < 10^6 \quad \Rightarrow \quad r + g \leq 999999. \quad (T)$$

We show $g \leq 998285$ by contradiction. Suppose $g \geq 998286$.

Case A: $r \leq 999$. Then by (D), $g \leq r^2 + 1 \leq 999^2 + 1 = 998002 < 998286$, contradicting the assumption $g \geq 998286$.

Case B: $r \geq 1000$. By (T), $r \leq 999999 - g \leq 999999 - 998286 = 1713$, so $1000 \leq r \leq 1713$ and $r^2 + 1 \leq 1713^2 + 1 = 2\,934\,370$. From (D), the cofactor satisfies $m = (r^2 + 1) / g \leq 2\,934\,370 / 998286 = 2.939\dots$, so $m \in \{1, 2, 3\}$. But $m = 3$ is impossible: a number $r^2 + 1$ is never divisible by 3, since squares are 0 or $1 \pmod{3}$, so $r^2 + 1 \equiv 1$ or $2 \pmod{3}$. Hence $m \in \{1, 2\}$.

Subcase $m = 1$: $g = r^2 + 1 \geq 998286 \Rightarrow r^2 \geq 998285 \Rightarrow r \geq 1000$. Then $r + g = r + r^2 + 1 \geq 1000 + 1000^2 + 1 = 1\,001\,001 > 999999$, contradicting (T).

Subcase $m = 2$: $r^2 + 1 = 2g$ is even, so r is odd. Also $g = (r^2 + 1)/2 \geq 998286 \Rightarrow r^2 \geq 1\,996\,571 \Rightarrow r \geq 1414$, and since r is odd, $r \geq 1415$. Then $r + g = r + (r^2 + 1)/2 = (r + 1)^2 / 2 \geq (1416)^2 / 2 = 2\,005\,056 / 2 = 1\,002\,528 > 999999$, again contradicting (T).

Every case is impossible, so the assumption $g \geq 998286$ fails. Therefore

$$g \leq 998285$$

for every admissible family with $k = 3$, and (with Part 2) for every admissible family with $k \geq 3$.

7 PART 4. THE BOUND 998285 IS ATTAINED

Take $k = 3$, $g = 998285$, and the residue $r = 1413$. Note

$$r^2 + 1 = 1413^2 + 1 = 1\,996\,569 + 1 = 1\,996\,570 = 2 * 998285,$$

so $1413^2 \equiv -1 \pmod{998285}$. Define the three distinct positive integers

$$\begin{aligned} a_1 &= 1413, \\ a_2 &= 1413 + g = 1413 + 998285 = 999698, \\ a_3 &= 1413 + 2g = 1413 + 1\,996\,570 = 1\,997\,983. \end{aligned}$$

They are distinct and positive, and

$$a_1 + a_2 + a_3 = 1413 + 999698 + 1\,997\,983 = 2\,999\,094 < 3\,000\,000,$$

so the family is admissible.

Since $a_1 \equiv a_2 \equiv a_3 \equiv 1413 \pmod{g}$, each $b_i = \text{product of the other two } a_j \equiv 1413^2 \equiv -1 \pmod{g}$, hence $g = 998285$ divides $b_i + 1$ for every i . Therefore the gcd of the three numbers $b_i + 1$ is a multiple of 998285. By the upper bound in Part 3 that gcd cannot exceed 998285, so it equals exactly 998285.

(Direct check: with $P = a_1 a_2 a_3$, one computes $\gcd(P/a_1 + 1, P/a_2 + 1, P/a_3 + 1) = 998285$, confirming the value. For the record, $998285 = 5 * 199657$, both factors being $\equiv 1 \pmod{4}$, which is consistent with -1 being a quadratic residue mod g .)

8 CONCLUSION

For every admissible choice the gcd g is at most 998285 (Parts 2 and 3), and this value is achieved (Part 4). Hence the maximum possible value of g is

$$g = 998285.$$